Executive Summary: Customer Churn & Retention Strategy

Strategic Brief for Executive Leadership & Banking Governance Committees

1. Core Problem & Portfolio Financial Exposure

Across 10,000 retail banking customer accounts in France, Germany, and Spain, portfolio attrition stands at **20.37%**. This represents **€185.6 million** in customer deposits currently lost to competitor banks. Rather than an evenly distributed attrition pattern, churn is heavily concentrated in two specific business risk clusters:

• **Regional Imbalance:** German accounts churn at **32.4%** — nearly double that of France (16.2%) and Spain (16.7%). German churners hold an average balance of €119,730, posing severe liquidity risks.

• **High-Value Demographic Churn:** Customers aged 46–60 experience an alarming **51.1%** attrition rate, depleting the bank's core wealth-accumulating customer base.

2. Customer Segment Overview & Churn Distribution

Segment Name	Clients	% Portf.	Avg Balance	Churn Rate	Strategic Priority
Affluent German Portfolio	1,873	18.7%	€129,849	33.2%	Tiered deposit rates & relationship managers
Mature Inactive Retainees	1,032	10.3%	€68,258	62.0%	Proactive outreach & service recovery
Multi-Product At-Risk	326	3.3%	€78,822	85.9%	Product fee audits & bundle waivers
Young Stable Core	3,115	31.2%	€35,062	4.4%	Cross-sell lending & digital expansion
Standard Retail Accounts	3,654	36.5%	€85,808	16.1%	Baseline digital satisfaction programs

3. Machine Learning Decision Engine Performance

A production-grade **Gradient Boosting Classifier** was deployed to replace static rules with real-time probability scoring:

• **High Discriminative Power:** ROC-AUC of **0.8670** and Accuracy of **86.85%**, outperforming baseline models.

• **Primary Risk Indicators:** Age (33.8%) and Number of Bank Products (25.1%) account for nearly 60% of total predictive signal, followed by Inactivity (10.6%) and German residence (6.6%).

4. Financial ROI Model & Retention Economics

Transitioning from reactive handling to proactive predictive retention delivers immediate financial return:

Financial Variable	Baseline Operational Metric	Projected Annual Impact
Targeted High-Risk Cohort	2,000 High-Probability Churn Accounts	Identified automatically by ML pipeline
Retention Offer Budget	€75 per customer (Waiver / Rate bonus)	€150,000 total investment
Retention Success Rate	25.0% target recovery	500 accounts successfully retained
Annual Gross Value per Retained Client	€850 net revenue margin / NIM	€425,000 stabilized revenue
Net Projected Annual Profit	€275,000 Net Benefit	183.3% Return on Investment (ROI)

5. High-Level Governance & Policy Implications

• **Algorithmic Fairness & Parity:** Under ECB algorithmic transparency mandates, models must be audited quarterly to ensure credit access and interest rates are not unfairly biased by protected demographic characteristics.

• **Product Bundling Governance:** The bank must reform product management to eliminate penalty fee stacking for customers holding 3+ products, which is currently driving an 86% churn rate.

• **Deployment:** Operationalize the live Streamlit dashboard across all European branch relationship managers to enable daily account triage.